

Learn About Designing Unilateral Low-Noise Amplifiers

In this article, we learn about noise parameters and use Smith charts to design a unilateral low-noise amplifier (LNA) for a specified gain.

Technical Article December 13, 2023 by [Dr. Steve Arar](#)

In receiver applications, the first amplifier in the signal chain has a dominant effect on the noise performance of the overall system. This amplifier should exhibit as low a noise figure as possible while providing an acceptably high power gain. The design procedure for this low-noise amplifier (LNA) should therefore account for both gain and noise performance.

In this article, we'll learn about how to design a unilateral LNA based on these requirements. We'll start off by exploring how the noise parameters of a two-port network are specified in RF applications, then work through the process of designing a unilateral amplifier that achieves both a specific gain and a specific level of noise. Finally, we'll test our design using the RF design software introduced in the [previous article](#) of this series.

Noise Parameters of a Two-Port Network

As my article on the [noise figure metric](#) discussed in detail, the output noise of a circuit depends greatly on its source impedance. Meanwhile, the noise factor (F) of a transistor connected to the source admittance $Y_S = G_S + jB_S$ is given by the following equation:

$$F = F_{min} + \frac{R_N}{G_S} |Y_S - Y_{opt}|^2$$

Equation 1.

where:

F_{min} is the minimum noise factor of the device

R_N is the equivalent noise resistance of the device

Y_{opt} is the optimum source admittance

G_S is the real part of the source admittance, Y_S .

We can see from this equation how F changes with the source admittance (Y_S). Observe that for $Y_S = Y_{opt}$, the noise factor reduces to its minimum value, F_{min} .

The quantities F_{min} , R_N , and Y_{opt} are called the noise parameters of the transistor. We don't calculate these—instead, they're either given by the manufacturer or obtained through measurement. F_{min} , which is sometimes given in dB as NF_{min} , changes with the transistor's bias point, temperature, and frequency of operation. The R_N parameter is a sensitivity factor, showing how fast the noise factor increases as the source admittance moves away from Y_{opt} .

At low frequencies Y_{opt} is real, but it becomes a complex value above 50 to 100 MHz for most active devices. For any given two-port network, we can find a value for Y_{opt} that minimizes the noise factor. Note that the S-parameters aren't present in Equation 1. In fact, the S-parameters of a device don't provide us with any information about its noise performance.

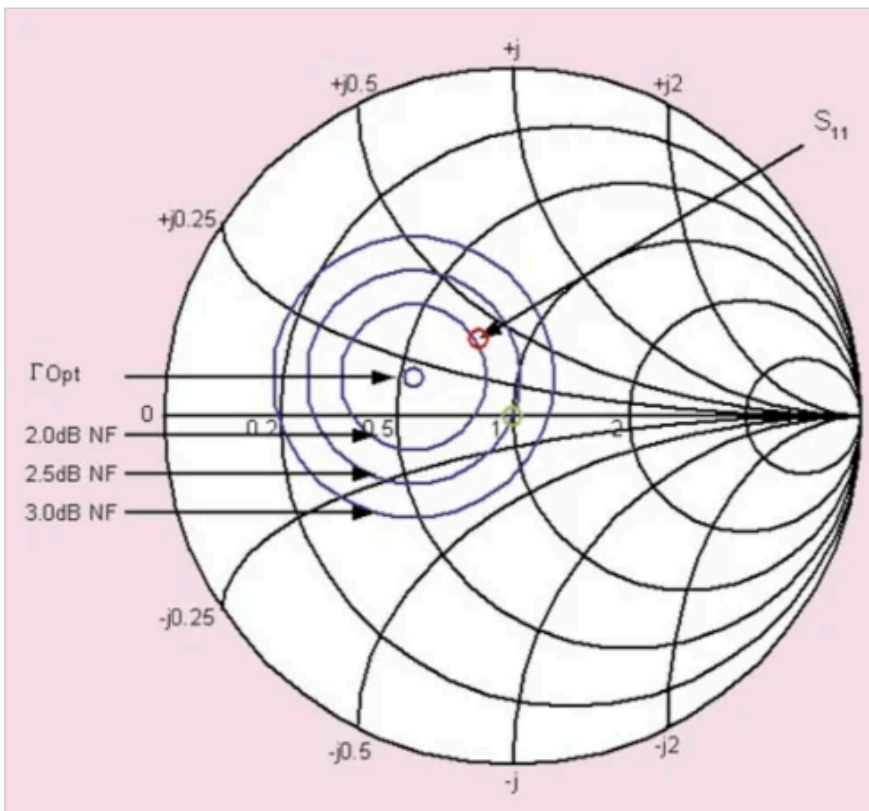
F , as stated previously, is the noise factor. It's expressed in linear terms. The noise *figure*, abbreviated as NF , is the noise factor converted to dB. The relationship between F and NF can therefore be expressed as follows:

$$NF = 10 \log_{10}(F)$$

Equation 2.

In practice, determining the NF dependence on the source impedance requires specialized noise measurement equipment. This equipment uses stub tuners to apply a range of complex impedances to the device, and these measurements are then analyzed to produce the contours of constant NF on the Γ_S plane.

Figure 1 shows the constant NF contours for a hypothetical device. As we'll shortly discuss in greater detail, these contours are circular in form.



*Figure 1. Smith chart displaying NF contours of a hypothetical device, demonstrating the impact of driving-point impedance on the noise figure.
Image used courtesy of [D. Boyd](#)*

Note that common noise figure analyzers and network analyzers are not capable of producing these NF contours.

Alternative Form of the Noise Factor Equation

The R_N parameter introduced above can also be specified as a conductance term, $G_N = \frac{1}{R_N}$. Also, instead of specifying the optimum admittance, it's possible to solve the equation by specifying either the equivalent optimum source impedance ($Z_{opt} = \frac{1}{Y_{opt}}$) or its associated optimum source reflection coefficient (Γ_{opt}). The parameters Y_{opt} and Γ_{opt} are related by the following equation:

$$Y_{opt} = \frac{1}{Z_0} \times \frac{1 - \Gamma_{opt}}{1 + \Gamma_{opt}}$$

Equation 3.

Using the Γ_{opt} parameter, Equation 1 can also be expressed as:

$$F = F_{min} + \frac{4R_N}{Z_0} \times \frac{|\Gamma_S - \Gamma_{opt}|^2}{(1 - |\Gamma_S|^2) \times |1 + \Gamma_{opt}|^2}$$

Equation 4.

Note that the amplifier's load reflection coefficient (Γ_L) doesn't appear in Equation 4. We can see from this that the output matching doesn't have any effect on the noise factor. However, a matched output can provide more gain and reduce the effect of the noise of the following stages.

There's generally a trade-off between the gain and noise performance of an amplifier—the minimum noise can't be achieved at the maximum gain.

Drawing Constant NF Circles

In order to draw the constant NF circles for a given noise factor (F), we first find the noise figure parameter (N). This is given by:

$$N = \frac{F - F_{min}}{4R_N/Z_0} |1 + \Gamma_{opt}|^2$$

Equation 5.

The center (c_F) of the constant NF circle is given by:

$$c_F = \frac{\Gamma_{opt}}{N + 1}$$

Equation 6.

and its radius (r_F), by:

$$r_F = \frac{\sqrt{N(N + 1 - |\Gamma_{opt}|^2)}}{N + 1}$$

Equation 7.

To cement these concepts, let's work through an example.

Example 1: Plotting Constant NF Circles

Assume that $Z_0 = 50 \Omega$ and $f = 1.4$ GHz for a transistor with the following S-parameters:

Table 1. S-parameters for an example transistor.

f (GHz)	S_{11}	S_{21}	S_{12}	S_{22}
1.4	$0.533 \angle 176.6$ degrees	$2.8 \angle 64.5$ degrees	$0.02 \angle 58.4$ degrees	$0.604 \angle -58.3$ degrees

The noise parameters of the device are:

$$NF_{min} = 1.6 \text{ dB}$$

$$\Gamma_{opt} = 0.5 \angle 130 \text{ degrees}$$

$$R_N = 20 \Omega.$$

Let's plot constant NF circles for this transistor at $NF = 2$ dB, 2.5, dB, and 3 dB. Table 2 summarizes the required calculations. Note that our equations use F , not NF , so we can't plug the noise figure values into the equations directly. Instead, we have to convert them from decibel measurements to the linear terms in which the noise factor is expressed.

Table 2. These calculation results allow us to plot constant NF circles for our example transistor.

NF	F	N	c_F	r_F
2.0 dB	1.58	0.05	$0.47 \angle 130$ degrees	0.20
2.5 dB	1.78	0.13	$0.44 \angle 130$ degrees	0.30
3.0 dB	2.00	0.21	$0.41 \angle 130$ degrees	0.37

These constant NF circles are plotted in Figure 2.

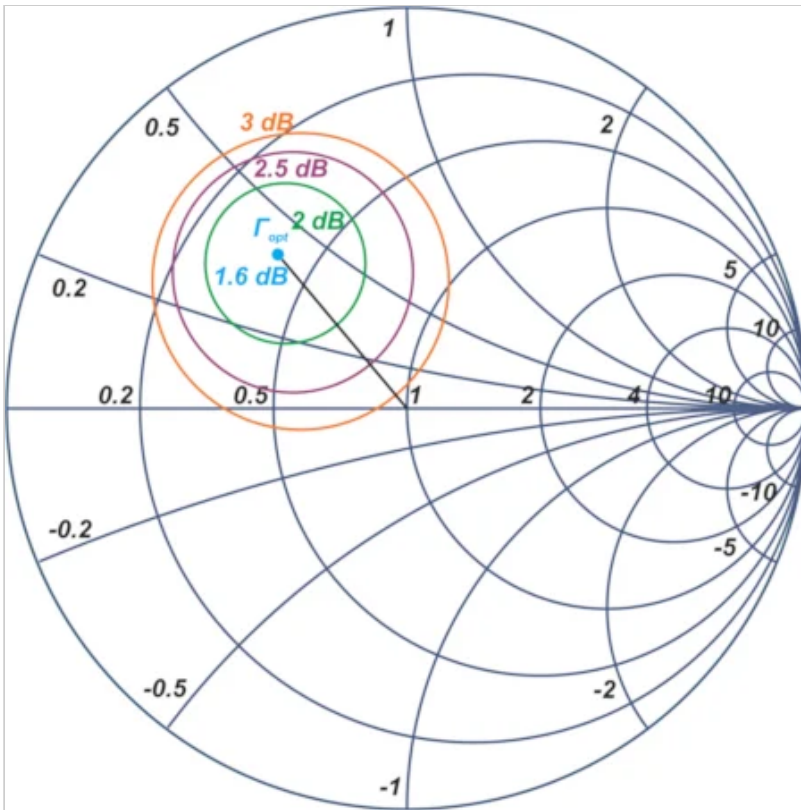


Figure 2. Constant NF circles for an example transistor. Image used courtesy of Steve Arar

Note that the centers of the constant noise circles lie on the line drawn from the center of the Smith chart to the point Γ_{opt} (see Equation 6). At Γ_{opt} , where we obtain $NF_{min} = 1.6$ dB, the noise circle transforms to a single point. As we increase the noise figure, the center of the circle moves closer to the origin, and its radius becomes larger.

Designing Unilateral RF Amplifiers for Both Gain and Noise

The constant NF circles are plotted in the Γ_S plane and can be used to find the appropriate source termination for a given noise figure. To account for both noise and gain, we need to plot the gain contours in the Γ_S plane as well. This is straightforward in the case of a unilateral device, where the gain of the input and output matching sections are independent of each other. We'll save the design of bilateral LNAs for the next article.

Example 2: Designing a Unilateral LNA for a Specific Gain and Noise Performance

Using the transistor from the previous example, let's design an amplifier that has a 2.5 dB noise figure with the maximum possible gain.

The transistor has a small S_{12} , suggesting that it might be considered unilateral. Applying the [unilateral figure of merit](#) (U), we obtain:

Scroll to continue with content

$$U = \frac{|S_{12}||S_{21}||S_{11}||S_{22}|}{(1 - |S_{11}|^2)(1 - |S_{22}|^2)} = 0.04$$

Equation 8.

Since U is less than 0.1, we immediately know that the error of the unilateral method is less than ± 1 dB. The unilateral approach can therefore be applied. We can also calculate the exact value of the error bound for the unilateral approximation. This works out to be:

$$-0.34 \text{ dB} < G_T - G_{TU} < 0.35 \text{ dB}$$

Equation 9.

This means that we should expect less than about ± 0.35 dB error in the actual gain of our final design.

Next, we determine $G_{S,max}$, the maximum possible gain for the input matching section of our unilateral device:

$$G_{S,max} = \frac{1}{1 - |S_{11}|^2} = 1.4$$

Equation 10.

which [converts to 1.46 dB](#).

This allows us to choose appropriate values for our constant gain circles. In this example, I arbitrarily chose to plot the $G_S = 0.5, 1, 1.28$, and 1.4 dB circles. The centers and radii of these constant G_S circles are given in Table 3.

Table 3. Centers and radii of constant G_S circles.

Gain	Normalized Gain	Center	Radius
$G_S = 0.50 \text{ dB}$	$g_S = 0.80$	$c_{S1} = 0.45 \angle -176.6 \text{ degrees}$	$r_{S1} = 0.34$
$G_S = 1.00 \text{ dB}$	$g_S = 0.90$	$c_{S2} = 0.49 \angle -176.6 \text{ degrees}$	$r_{S2} = 0.23$
$G_S = 1.28 \text{ dB}$	$g_S = 0.96$	$c_{S3} = 0.52 \angle -176.6 \text{ degrees}$	$r_{S3} = 0.15$
$G_S = 1.40 \text{ dB}$	$g_S = 0.99$	$c_{S4} = 0.53 \angle -176.6 \text{ degrees}$	$r_{S4} = 0.07$

Figure 3 plots these circles and the $NF = 2.5 \text{ dB}$ circle in the Γ_S plane.

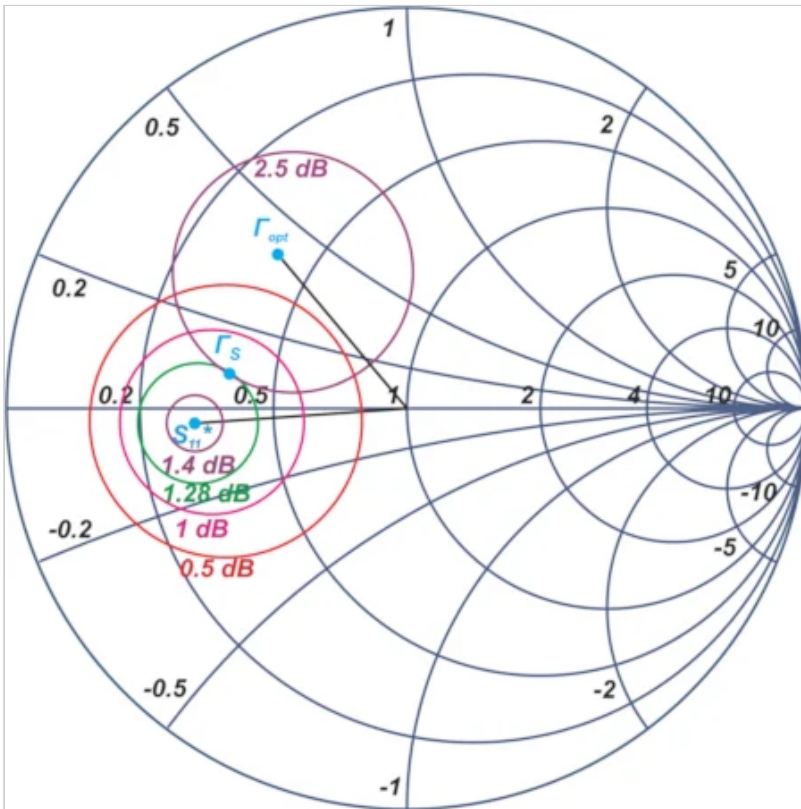


Figure 3. Constant G_S and NF circles in the Γ_S plane. Image used courtesy of Steve Arar

The $G_S = 1.28 \text{ dB}$ gain circle only intersects the noise circle of $NF = 2.5 \text{ dB}$ at $\Gamma_S = 0.45 \angle 169.17 \text{ degrees}$. Any higher value of G_S will move us away from Γ_{opt} , resulting in a larger noise figure.

For the output section, we choose a conjugate match to maximize the gain. This results in:

$$\Gamma_L = S_{22}^* = 0.604 \angle 58.3 \text{ degrees}$$

Equation 11.

and:

$$G_L = \frac{1}{1 - |S_{22}|^2} = 1.57$$

Equation 12.

which converts to 1.96 dB.

The total gain is calculated as:

$$G_{TU} = G_S + G_0 + G_L = 1.28 \text{ dB} + 8.94 \text{ dB} + 1.96 \text{ dB} = 12.18 \text{ dB}$$

Equation 13.

$G_0 = |S_{21}|^2$ in the above equation. This is the basic Z_0 -based transducer power gain of the transistor.

Next, we use a Z Smith chart to design the input and output matching networks. For the input matching section, we locate Γ_S on the Smith chart in Figure 4 and find its associated normalized admittance (y_S) through a 180 degree rotation along the constant $|\Gamma_S|$ circle.

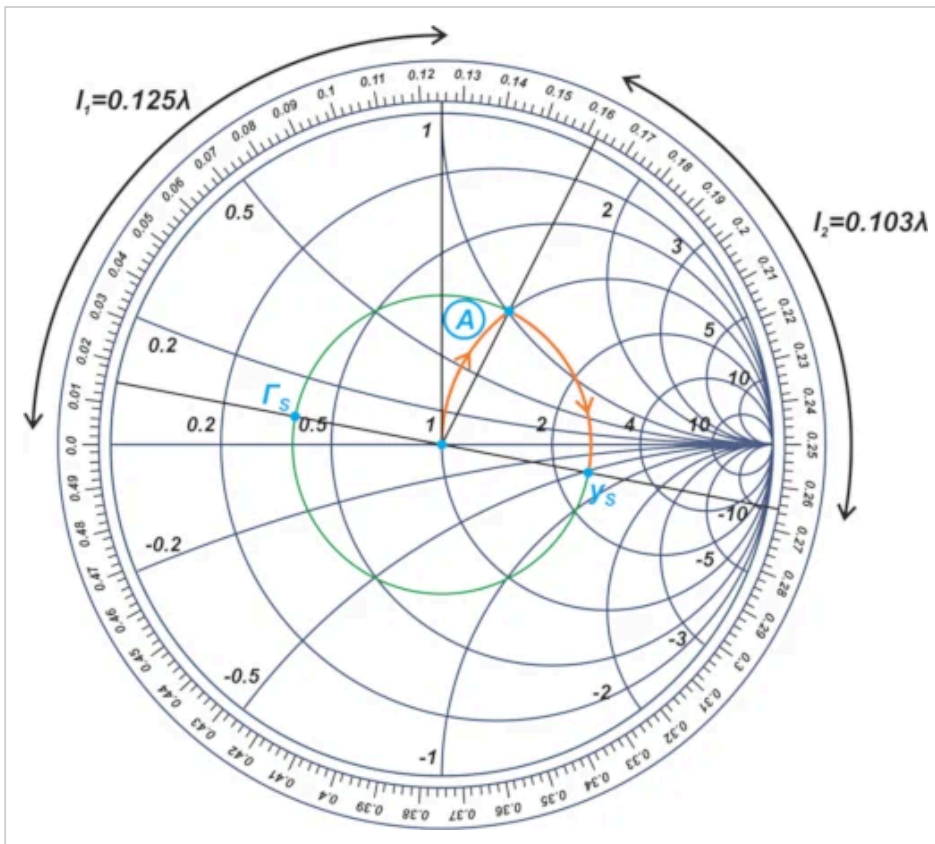


Figure 4. The constant $|\Gamma_S|$ circle. Important points on the circle are marked in blue. Image used courtesy of Steve Arar

From now on, we interpret the Smith chart as a Y Smith chart. We want a circuit that takes us from the center of the chart, located at the $50\ \Omega$ termination, to y_S . The intersection point of the constant $|\Gamma_S|$ circle with the $1 + jb$ circle is marked as point A , which has a susceptance of approximately $j1$.

When designing the input matching section of the two-port network, we add a parallel open-circuited stub of length $l_1 = 0.125\lambda$ to the $50\ \Omega$ termination to create a susceptance of $j1$. We then add a series line of length $l_2 = 0.103\lambda$ to travel along the constant $|\Gamma_S|$ circle to y_S .

The output matching section can be designed in a similar way. As Figure 5 shows, the output matching network needs an open-circuited stub of length $l_3 = 0.157\lambda$ and a series line of length $l_4 = 0.243\lambda$.

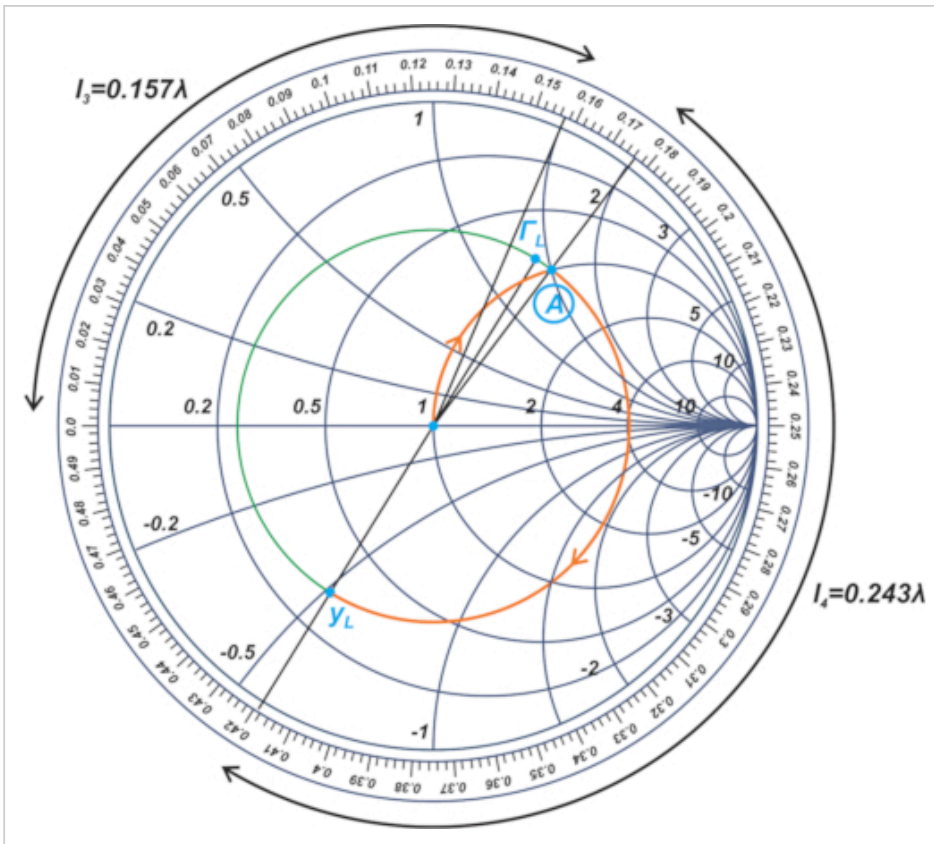


Figure 5. Constant $|\Gamma_L|$ circle. Note the values of l_3 and l_4 . Image used courtesy of Steve Arar

Figure 6 shows the AC schematic of the final amplifier design.

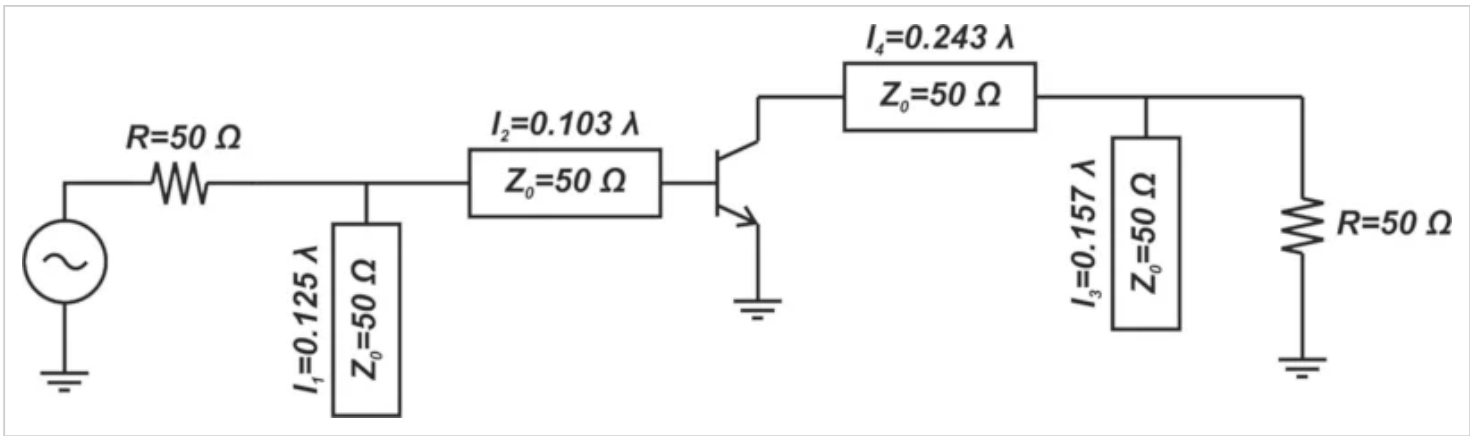


Figure 6. Final design of the example LNA. Image used courtesy of Steve Arar

Now we can use design software to verify the performance of the circuit.

Adding Noise Parameters to a Touchstone File

As we learned in the recent article about [RF amplifier stabilization techniques](#), the Touchstone (.s2p) file format is commonly used in RF design software to specify the S-parameters of a two-port network. Table 4 shows the .s2p file for the S-parameters of the amplifier in Figure 6. The noise parameters are also included at the end of the file, though this is optional.

Table 4. S-parameter and noise parameters of the amplifier in Figure 6, saved as a Touchstone file.

```
# MHz S MA R 50
! freq magS11 angS11 magS21 angS21 magS12 angS12 magS22 angS22

1400 0.533 176.6 2.8 64.5 0.02 58.4 0.604 -58.3

! Noise Parameters
1400 1.6 0.5 130 0.4
```

Recall that the options line, which starts with the # mark, contains the header information. This header information specifies the frequency unit and the data format for the S-parameters. The term “R 50” in the options line indicates that the load termination resistor for the S-parameters is 50 Ω. The lines that start with the ! symbol are comment lines.

As you can see, there isn’t a separate options line for noise parameters. In order for the simulator to be able to distinguish where the S-parameters data ends and the noise data begins, the first frequency of the noise parameters must be less than or equal to the highest frequency of the S-parameters.

The data format for the noise information is as follows:

- The first column specifies the frequency (1400 MHz).
- The second column gives the minimum noise figure in dB (1.6 dB).
- The next two columns give the magnitude and phase of the optimum reflection coefficient ($\Gamma_{opt} = 0.5 \angle 130$ degrees).
- The last column is the effective noise resistance ($R_N = 20 \Omega$) normalized to the system impedance we defined in the options line.

Linking the above .s2p file to a s2p component in Pathwave ADS, we can analyze the gain and noise performance of the system. The Pathwave ADS schematic we produce is shown in Figure 7.

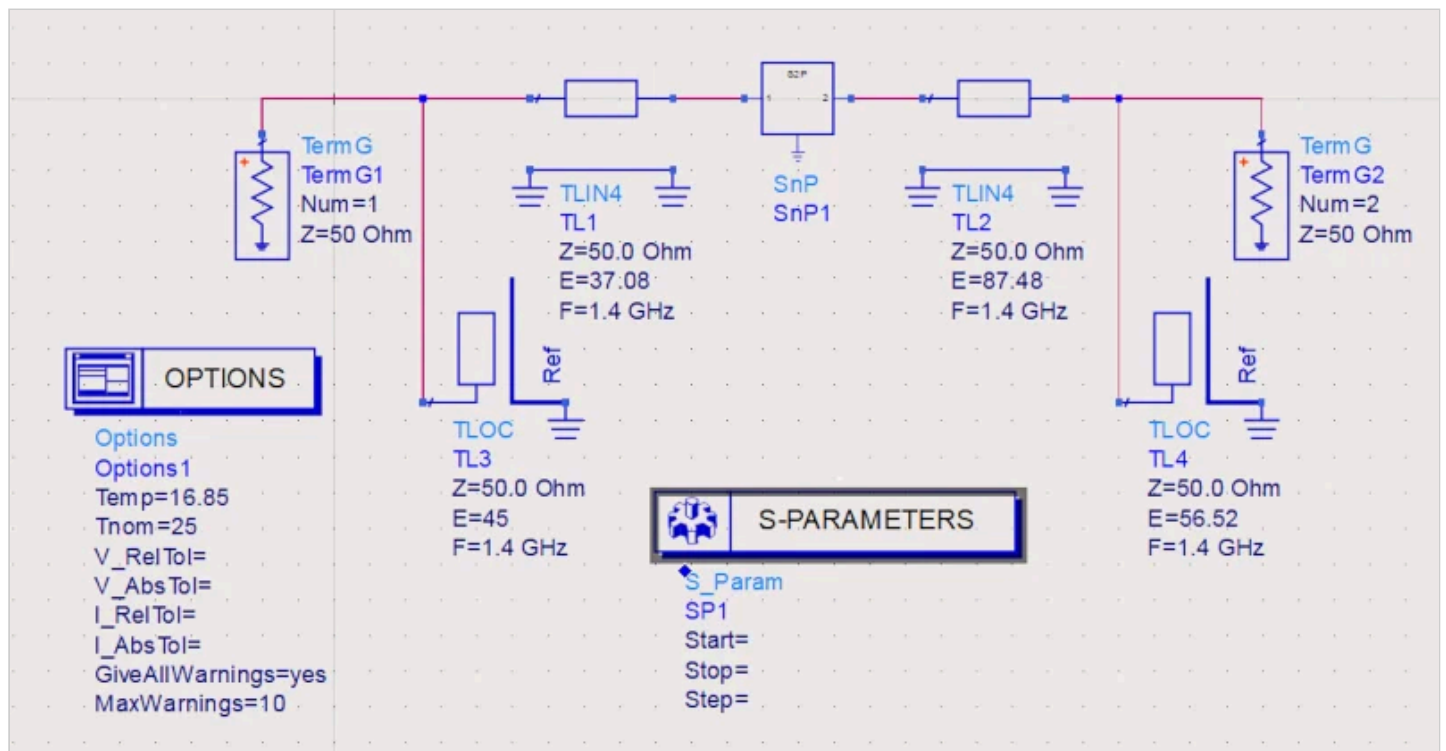


Figure 7. Pathwave ADS schematic of an example amplifier. Image used courtesy of Steve Arar

Note that the simulation temperature is set to 16.85 °C, ensuring that the noise figure measurement is consistent with the IEEE [definition of the noise figure](#). The computer analysis shows that the circuit we've designed has a gain of 12.466 dB and noise figure of 2.522 dB. These numbers are acceptably close to our design specifications.

Wrapping Up

The latter part of this article focused on working through examples. Here's a brief summary of the concepts introduced earlier on, should you want to review them:

- The noise performance of a two-port network with given operating conditions can be fully characterized by its noise parameters: F_{min} (or NF_{min}), Γ_{opt} , and R_N .
- The constant NF circles are plotted in the Γ_S plane and can be used to find the appropriate source termination for a given noise figure.
- To account for both noise and gain, we need to plot the gain contours in the Γ_S plane as well. This is straightforward if the device is unilateral.

Note that it's less straightforward to plot gain contours on the Γ_S plane if the device is bilateral. We previously used [operating power gain](#) (G_P) circles to design a bilateral amplifier for a specific gain, but G_P circles are in the Γ_L plane and don't directly specify the usable source terminations.

Fortunately, there's a method based on the available power gain (G_A) concept that allows us to plot the gain contours of a bilateral device in the Γ_S plane. We'll discuss how to use constant G_A circles to design bilateral amplifiers for both gain and noise performance in the next article.

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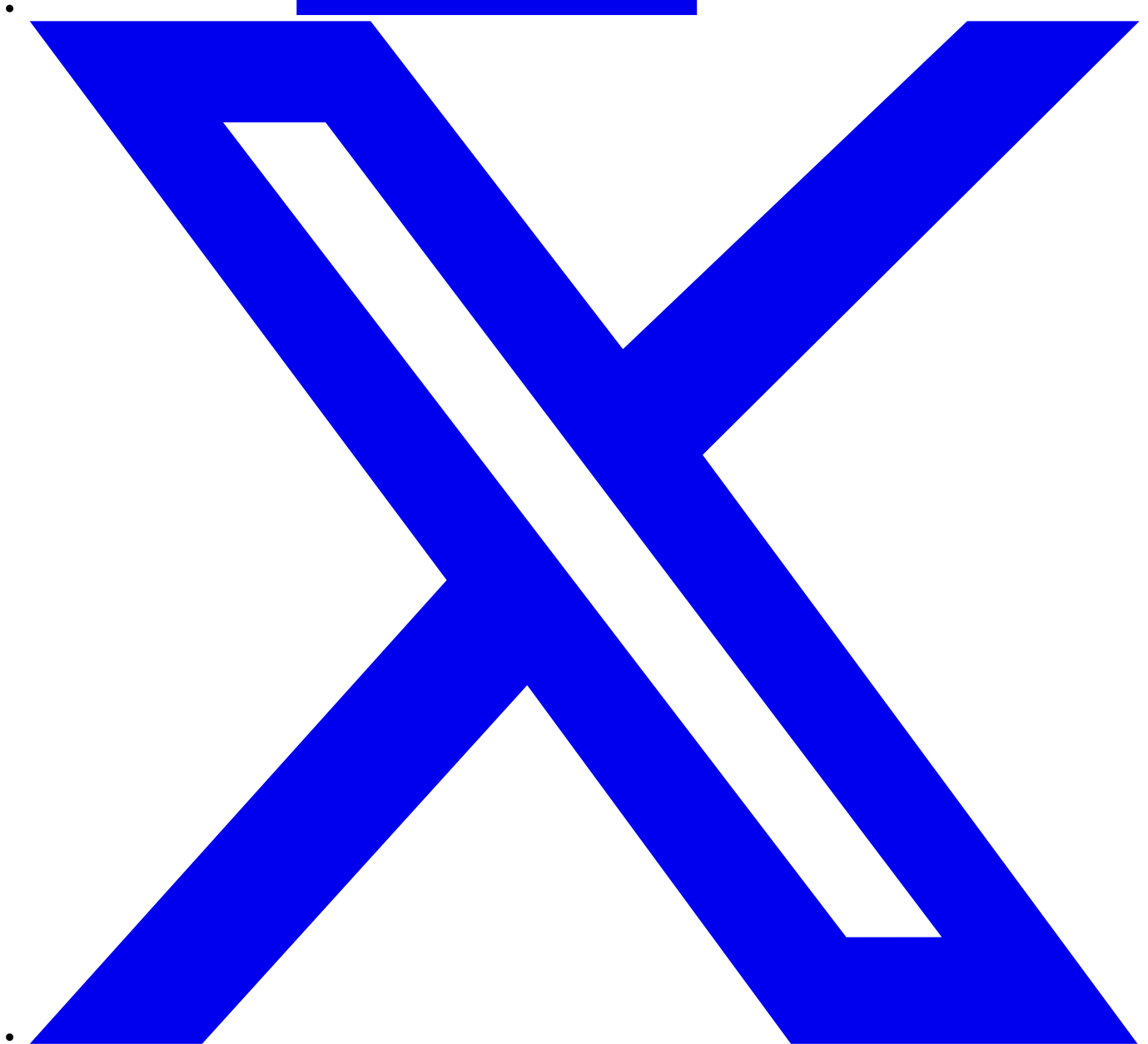
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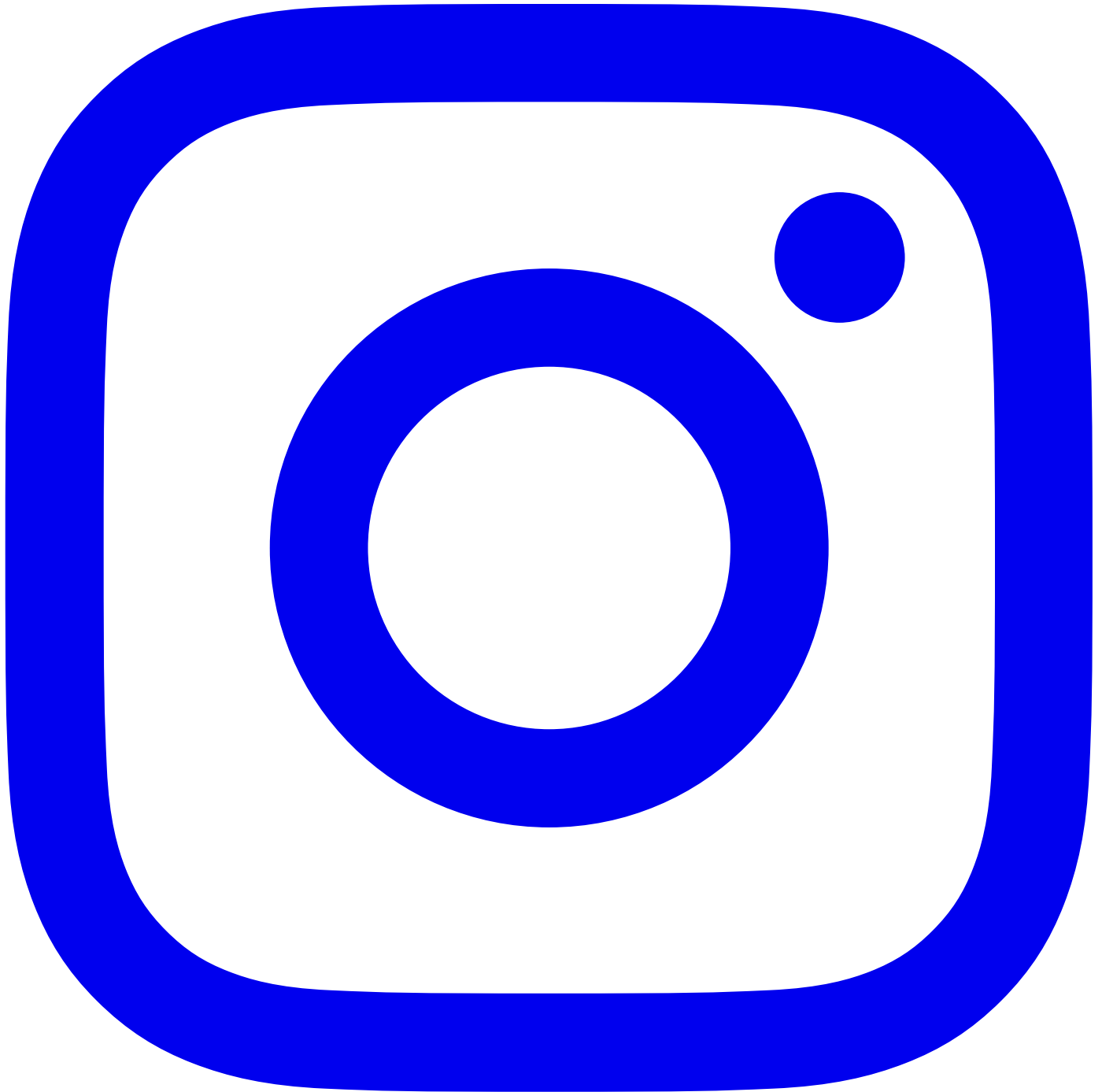
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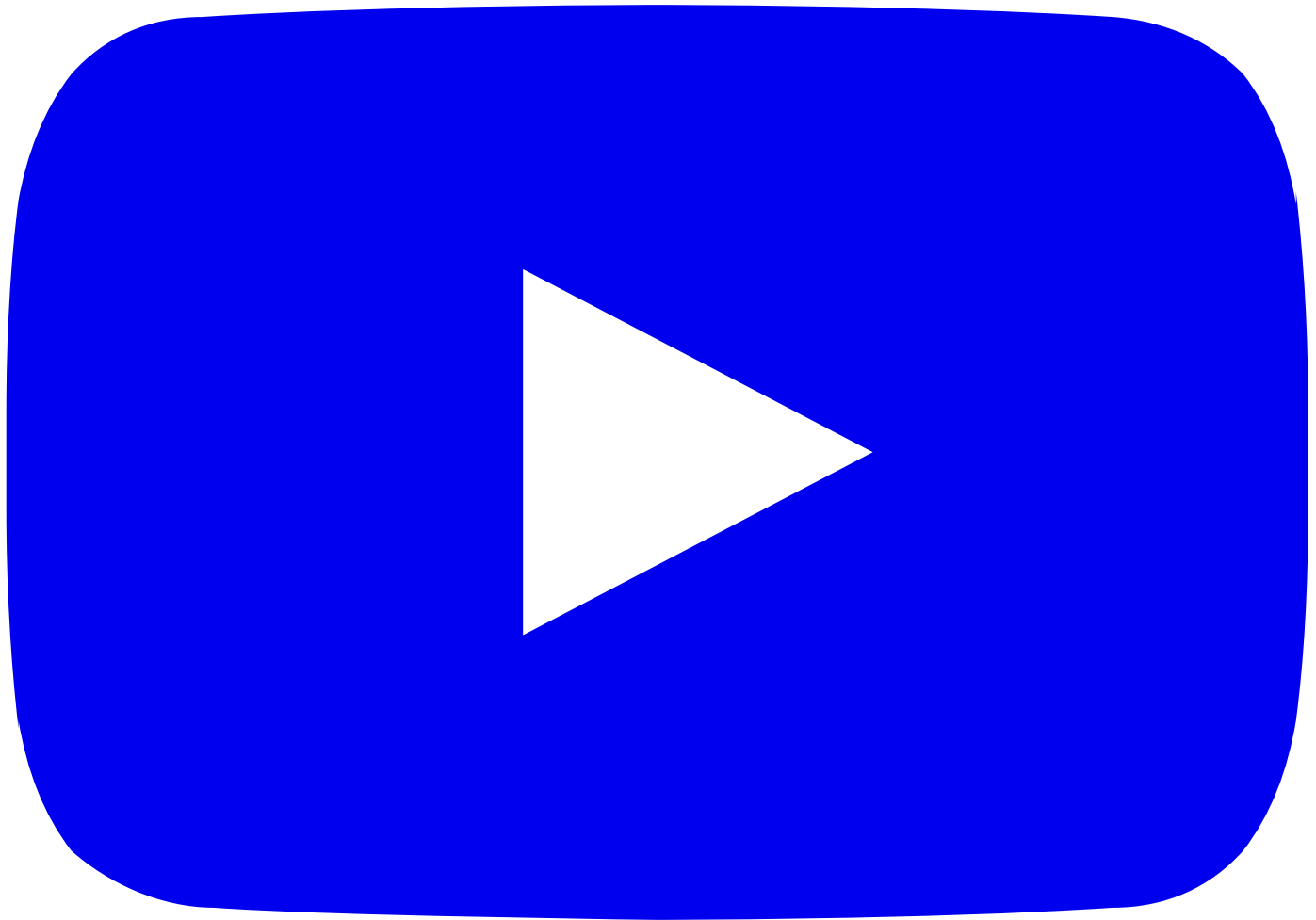
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